



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Let's learn about computer architecture!



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.
Make the information system design report, based on the selection of both hardware and software tools required for the IT solution.

Can Do Statements:

- I can explain what computer architecture is.
- I can discuss the roles of computer architecture.
- I can give detailed information related to computer architecture.

Ask your partner:

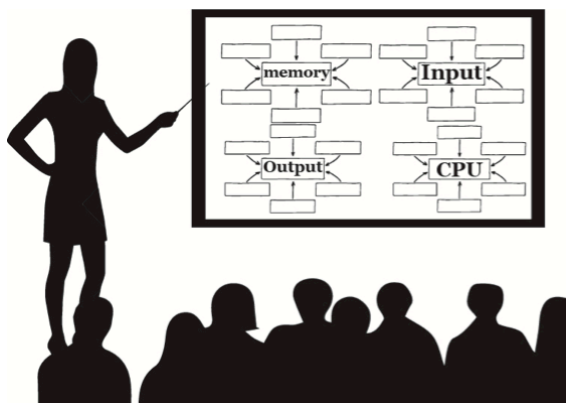
- What do you think is more important: hardware or software?"
- Why do you imagine the concept of computer architecture is important in computing?





Let's Explore!

There's a computer architecture conference today!



Instructions: Read the text below.

Speaker: Good evening, we are going to talk about computer architecture. So, let's start by explaining this concept. Computer architecture refers to some methods and rules that define computer systems, in terms of their functions, as well as their organization and implementation. It is a representation of the design requirements for certain parts of the computer, especially the CPU and its internal connection with other parts. For that reason, computer architecture is as important as software architecture and, for some, it is even considered more important. Some examples of computer architecture evolution are MS-DOS operating system, Apple's Lisa, and Microsoft Windows.

Now, let me give you more details about computer architecture. It works through pipeline processors, which consist of linear and sequential segments that perform specific tasks. For instance, the external data are processed by the system and, then, it makes operations with them and with the memory-stored data. Finally, it produces new data for external use. This process can be more complex than others since it involves microarchitecture design, logic design, instruction set architecture design, and implementation.

Attendee: Can you explain more about those roles in computer architecture?

Speaker: Sure, Instruction set architecture (ISA) has the role of an interface between the hardware and the software of the computer. The microarchitecture design improves the software performance by implementing processors that transfer data in a faster way. Finally, though implementation isn't part of the computer architecture field at all, it is as essential as the previous ones since it implies checking the transistors design, creating physical circuits, and validating the final design.

Attendee: What do you mean by validating the final design?

Speaker: It involves checking the functionality of the complete process.



Let's Practice!

Slides About What Computer Architecture Is.

Instructions: Organize the mystery sentence.

1. rules / methods / computer / Specific / architecture. / and / define

2. in / Data / stored / are / addresses. / memory / specific

3. internal / determines / work. / the / Architecture / computer / parts / how

4. computer. / organization / Microarchitecture / establishes / of / the / the



Let's Practice!

The History of Computers.

Instructions: Match the two parts of each sentence.

COLUMN A

1. Computers make complex
2. Computers
3. Thanks to the evolution of memory,
4. Operative systems give computers
5. Nowadays, computer microarchitecture

COLUMN B

- a. computers store more data now than before.
- b. allows having smaller pocket computers.
- c. calculate formulas.
- d. and simple everyday tasks faster.
- e. more dynamic functions.



Extension Activity

The Main Components of Computer Architecture.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Research about the main steps or components of computer architecture.
- Make a diagram presenting each part and how they connect to one another.
- Show your diagram to a partner. Explain and answer any questions about the topic.

Self-Assessment

Yes No

1. I research about computer architecture.

☐ ☐

2. I make a diagram to explain.

☐ ☐

3. I present my diagram to a partner.

☐ ☐


Target Words

- architecture** (noun): the internal design of hardware and software in a computer
 e.g. Computer architecture refers to some methods and rules that define computer systems, in terms of their functions, organization, and implementation.
- method** (noun): a procedure to do something
 e.g. Computer architecture refers to some methods and rules that define computer systems.
- pipeline processor** (noun): a sequence of processed data in which the output of one factor is the input of the following factor
 e.g. It works through pipeline processors, which consist of linear and sequential segments that perform specific tasks.
- microarchitecture** (noun): the organization of the computer; for instance, the implementation of processors
 e.g. It involves microarchitecture design, logic design, instruction set architecture design and implementation.
- circuit** (noun): an electronic path that communicates one point with another
 e.g. It implies to check the transistors design to create physical circuits and to validate the final design.
- validate** (verb): to confirm something
 e.g. What do you mean by validating the final design?



Learning Points

Adjectives As... As, Er, and More.

When you need to compare the different characteristics of something, you can use **comparatives**.

There are different ways to compare characteristics of objects. This learning point helps you understand three of them. The first one is by adding **-er** to short adjectives. The second form is by using the structure **more...than** with long adjectives. The last one is by using **as...as** to state that there is some equality between the two objects you compare.

The following chart exemplifies the appropriate use of these three cases of comparison.

1. -er (short adjectives)	adjective + er + than + Complement Example: This program runs faster than the default application.
2. more...than (long adjectives)	more + adjective + than + complement Example: The previous design is more interesting than the new one.
3. as...as (any adjective)	as + adjective + as + complement Example: In computer architecture, logic design is as fundamental as microarchitecture design.

Examples

- As you can see computer architecture is as important as software architecture and, for some, it is even considered more important.*
- This process can be more complex than other ones since it involves microarchitecture design, logic design, instruction set architecture design, and implementation.*
- Though implementation isn't part of the computer architecture field at all, it is as essential as the previous ones.*





Learning Points

Timeline: the Evolution of Personal Computers.

The personal computer, or PC, is one of the greatest inventions of the XX century. Thanks to this technological tool, we simplify all of our activities, from daily tasks to the most complex processes in our jobs.

Take a look at the following timeline to see the progress of personal computers through the years:

